

Time-calibrated Desmognathus phylogeny with audited bootstrap trait distributions

Size panel: all 21 species, with 50 manually vetted largest cells and corresponding nuclei each. Genome panel: 20 common-support species (relative IOD, not pg); D. ochrophaeus is shown without an IOD violin because it has limited image-quality overlap. No phylogenetic fills.

Descriptive species-level Spearman ρ : IOD–nucleus 0.45; IOD–cell 0.55; nucleus–cell 0.56. These correlations are not phylogenetically corrected.

Time-calibrated phylogeny

Relative nuclear-IOD

Nucleus area

Cell area

